FRISCO ISD ROBOTICS / BRADLEY SHOGREN

Robotics that protect people

A firefighter’s perspective. A few real examples. Better questions.

What could a robot do in the first 60 seconds to help someone make a safer decision?

PROTOTYPE + TESTING

See before entering

NIST’s ResponseWing project combined drone flight with onboard 3D mapping for search and rescue. A scene map can help responders understand where to look before committing a crew.

Ask: What information would actually change your next move?

Explore: NIST ResponseWing

REAL-WORLD STUDY

Bring help faster

In Sweden, drone-delivered AEDs were used before ambulances arrived in a real-world cardiac-arrest study. An automated external defibrillator still needs someone at the scene to use it.

Ask: How do you connect dispatch, delivery, and a willing bystander?

Explore: Karolinska AED research

DEPLOYED EXAMPLE

Take the risky position

The Los Angeles Fire Department uses the RS3, a remotely operated firefighting vehicle. It puts water and a camera where sending a person may be dangerous. Remote control is valuable robotics, too.

Ask: When is human control better than adding autonomy?

Explore: LAFD RS3

ENGINEERING CHALLENGE

Explore the unseen

A small robot could inspect rubble or carry sensors toward a hazardous area. NIST tests responder robots for the mobility, manipulation, and communication these missions demand.

Ask: Can it return a useful location, not just an interesting image?

Explore: NIST response-robot tests

THE ROBOT IS ONLY PART OF THE SYSTEM

SENSE — Collect useful data → **INTERPRET** — Find what matters → **DECIDE** — Keep people in control → **ACT** — Make the response safer

Where AI might help: highlight patterns in images, build maps, or prioritize alerts. The examples above do not all require AI. Start with the mission, then choose the technology.

Put every idea through the field test: Smoke? Water? Rubble? Weak signal? Gloves? A small department’s budget? And what happens when it fails?

Conversation aid, not a presentation script. Pick one example and follow the curiosity.